

On the cyclic encoding of control₂

Semantic Control Systems (1995), Definition 28

Claude (Fable 5.1), as prompted by Shingai Thornton*
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Source. C. Joslyn, “Semantic Control Systems”, *World Futures* 45 (1995) 87–123, cited below as J95 with the journal’s page numbers. Definition 28 (p. 103): “A control₁ system $CS = \langle C, O \rangle$ is also a control₂ system if the variation in the constraint₁ on O , induced by C ’s variation, is itself constrained₂.” Proposition 29 (p. 106) splits the object of control, $O = \langle O_E, O_I \rangle$, giving

$$CS = \langle C, O = \langle O_E, O_I \rangle \rangle \quad (30)$$

and §3.5 (p. 107) names the relations among the three components (Figure 1): “Thus together O_E and O_I form an afferent-efferent loop.”

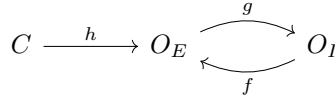
This note records the three named relations of Figure 1 as the arrows of a quiver and takes the free category on it (§1); a theorem follows from the loop alone (§2); §3 says what it does not claim; §4 asks whether the drawing is faithful to what you meant; §5 is a separate provenance item.

1 The encoding

Definition 1 (the quiver Q_J). *Vertices* $\{C, O_E, O_I\}$. *Generating arrows*

$$h: C \rightarrow O_E, \quad g: O_E \rightarrow O_I, \quad f: O_I \rightarrow O_E.$$

Convention. *Arrows point in the functional direction of Figure 1* (p. 107), $x \rightarrow y$ meaning “ x acts on y ”: C disturbs O_E ; O_E constrains O_I ; O_I feeds back to O_E .



symbol	role in Q_J	J95 (page)
C	vertex	“the global environment” (107); “a source of variation to the system O ” (107)
O_E	vertex	“some efferent component of O , the mechanism whose actions assert compensating variation on O_I ” (107)
O_I	vertex	“the controlled variables” (107)
h	arrow	“The variation of C affects O through the relation h ” (106)
g	arrow	“the final constraint ₁ of O_E on O_I ” (106)
f	arrow	“whatever relation there might be back from O_I to O_E ” (106)

Definition 2 (the shape). $\mathcal{I}_J := \mathbf{Paths}(Q_J)$, the free category on Q_J : objects are the vertices, morphisms are finite directed paths, composition is concatenation, identities are empty paths. No relations are imposed.¹

Remark 1 (what a quiver forgets). Q_J records only that a named dependency exists. It does not carry: (a) that f, g, h are functions rather than relations (§4.3, p. 114); (b) the asymmetry of the loop (§4.3, p. 114: f and g “are not thereby symmetric or complementary”); (c) the hierarchy of C over O and the “large loop gain” (p. 108); (d) the second loop. §3.7.2 (p. 110) says “there is also a kind of loop between C and O_E ” by reciprocal variation, and §3.7.1 (p. 109) rules it one-way at the level of action (“ O_E mirror the actions of C , but not vice versa”). Q_J follows §3.7.1: h is a single arrow. (e) Turchin’s alternative, in which g “flows only through the environment” (p. 108). Items (d) and (e) are encoding forks, not oversights; each yields a different Q .

2 The theorem

Write $\ell := f \circ g \in \text{Hom}_{\mathcal{I}_J}(O_E, O_E)$ for the feedback loop, a path of length 2.²

Lemma 1. *For $n \in \mathbb{N}$, ℓ^n has length $2n$; hence $n \mapsto \ell^n$ is injective and $\text{Hom}_{\mathcal{I}_J}(O_E, O_E)$ is infinite.³*

Proof. Length is additive under concatenation, so $|\ell^n| = n|\ell| = 2n$. Distinct n give distinct lengths, hence distinct paths. $\square$

Theorem 1. *Let $\mathcal{C}$ be any category in which every hom-set is finite. No functor $F: \mathcal{I}_J \rightarrow \mathcal{C}$ is faithful.⁴*

*Byline per the Multivar repository’s convention. The encoding decision, the readings of J95, and §4 are Shingai Thornton’s; §2 is machine-checked in github.com/halcyonic-systems/systems-science-foundations, files `Systems/Category/{ShapeJoslyn, CyclicObstruction, JoslynIncomparability}.lean`, commit 4288793.

¹`JoslynShape := Paths JoslynPosition, ShapeJoslyn.lean`. `Paths` and `Path.length.comp` are Mathlib’s.

²`joslynLoop, joslynLoop.length : joslynLoop.length = 2 := rfl`.

³`cyclePow.length, cyclePow.injective, infinite_path_of_cycle`, all proved for an arbitrary quiver and an arbitrary positive-length loop.

⁴`joslyn.no_faithful_functor`, instance of the general `paths.no_faithful_functor_of_cycle`. Axioms: `propext`, `Classical.choice`, `Quot.sound`. No sorry.

Proof. A faithful functor is injective on each hom-set. By Lemma 1 it would inject the infinite set $\text{Hom}(O_E, O_E)$ into the finite set $\text{Hom}(FO_E, FO_E)$. $\square$

Corollary 1. *Let $\mathcal{I}_K := \text{Paths}(R \rightarrow T)$ be the shape of Klir’s $S = (T, R)$: two objects, one generating arrow. Every hom-set of $\mathcal{I}_K$ is a subsingleton, hence finite. There is no faithful functor $\mathcal{I}_J \rightarrow \mathcal{I}_K$.⁵*

Remark 2 (why this is the boundary and not a curiosity). Every other tradition encoded in the same program (Klir, Bunge, Mobus, Myers, Wymore, Mesarović) has an acyclic Q , hence finite hom-sets, and is placed by faithful comparison functors; Q_J is the first cyclic shape (Spivak’s value loop is the other). Feedback is where the finite-shape method stops, and Theorem 1’s sole premise is the cycle.

3 What the theorem does not say

1. **It is presentation-relative.** From the theorem’s own docstring, written before anyone asked you: “both shapes are encodings of primary texts as dependency quivers, and a different defensible encoding of Joslyn 1995 — one that does not make the control loop a cycle at quiver level — could dissolve the obstruction. Not a maximality result.”
2. **It is about Paths, which imposes no relations.** Impose one and it can die. If feedback is taken to be idempotent, $\ell \circ \ell = \ell$, then $\text{Hom}(O_E, O_E)$ collapses to $\{\text{id}, \ell\}$ and Theorem 1 fails. Whether `control2` licenses that identification is a claim about your text, not about category theory.
3. **Re-bracketing does not dissolve it.** Your (31) $\langle O_E, \langle C, O_I \rangle \rangle$ and (32) $\langle O_I, \langle C, O_E \rangle \rangle$ regroup the vertices; neither removes an arrow, so the loop and the theorem survive both. (32) adds a second loop if the “kind of loop” of §3.7.2 is drawn, which only strengthens the obstruction.
4. **Unrolling would dissolve it, but your text disfavors it.** Index the loop by level: vertices $C, O_E^{(k)}, O_I^{(k)}$ for $k \in \mathbb{N}$; arrows $h: C \rightarrow O_E^{(0)}, g_k: O_E^{(k)} \rightarrow O_I^{(k)}, f_k: O_I^{(k)} \rightarrow O_E^{(k+1)}$. Acyclic, every hom-set finite, obstruction gone (stated, not machine-checked). But Proposition 29 is introduced as avoiding “the logical regression” (p. 106), and repeated passes are “the familiar cycling of O_I around its optimal state” with residual error “resubmitted to the control mechanism via the feedback loop” (p. 112). Passes are dynamics within one level, not new levels.
5. **It does not say `control2` is incomparable in principle.** Only that the method placing the six acyclic traditions cannot reach a cyclic one; a quotient, enrichment, or 2-categorical treatment of the loop is neither ruled out nor attempted.

4 The question, which only you can answer

In the margin of *Computational Systems Ontology: Towards Practical Integration of the Systems Sciences with BERT* (Thornton, August 2025, p. 4) you wrote: “One could argue that control is always so circular.” The encoding took that literally: one system, one level, a cycle at quiver level. Your text supports that much: the hierarchy is between C and O (p. 104), O_E and O_I are “more tightly coupled to each other than either is to C ” (p. 107), the regress is avoided (p. 106), and repetition is cycling (p. 112). So Q_J ’s hierarchy is carried by h alone and the f, g loop is level-flat. Theorem 1 needs one further thing, and it is the one thing your text does not state: *that ℓ^n is a different morphism for each n* . The free category makes it so by construction. Your dynamics treats the n -th pass as the same feedback relation applied again, converging on an attractor. The question is which of these `control2` is: a loop under free composition, where every power is a distinct control trajectory, or a single feedback relation whose iteration is dynamics rather than new morphisms, so that a relation on ℓ (§3 item 2) belongs in the encoding and the obstruction disappears. Four outcomes, each a result:

1. **Free composition.** Every ℓ^n is a distinct trajectory; Theorem 1 becomes author-endorsed.
2. **A relation on the loop.** You state which (idempotence, a fixed-point law, a single endomorphism with dynamics as its action); $\text{Hom}(O_E, O_E)$ becomes finite for that encoding, the obstruction dissolves, and presentation-relativity is exhibited rather than asserted.
3. **An acyclic encoding** after all, unrolled or otherwise, against the reading of pp. 106 and 112 above; then the reading is what needs correcting.
4. **None of these.** The quiver is the wrong instrument for the question, or something not anticipated here. Also a result, and likely the most informative one.

5 One provenance item, separately

Definition 28 is entry 009 of the definition atlas at `math.systems`, page-image transcribed and hand-checked, with five of your case rulings (thermostat and the decoupled-envelope pair admitted; damped spring, boiling liquid, spinning top refused). One grading is offered for correction: Definition 25 (`control1`) is introduced on p. 101 as Turchin’s 1992 definition, “we offer it again here,” so it is graded a restatement of Turchin and Definition 28 is recorded as the first control definition that is yours alone.

⁵`no_faithful_joslyn_to_klir`; the finiteness hypothesis is discharged by an instance built from `klirHomSubsingleton`, not by inspection in prose.